# Nikhil Ram Mohan, Ph.D.

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## PROFESSIONAL SUMMARY

Translational scientist with 18+ years of experience leading multidisciplinary research programs that integrate computational biology, microbiology, molecular biology, immunology, and clinically annotated patient cohorts to advance biomarker discovery, molecular diagnostics, and translational medicine. Recognized for bridging experimental biology, quantitative analytics, and clinical investigation to develop rigorous scientific strategies, evaluate emerging technologies, and accelerate collaborative research across academia and industry. Extensive experience directing clinical biospecimen research, developing computational methodologies, mentoring scientists, and translating complex biological questions into clinically meaningful research with applications in diagnostic and medical technology development.

## CORE EXPERTISE

Translational Research Strategy | Biomarker Discovery | Clinical Study Design | Molecular Diagnostics | Medical Technology Evaluation | Clinical Biobanks | Computational Biology | Quantitative Image Analysis | Multi-Omics Integration | Scientific Program Leadership | Cross-Functional Collaboration | Industry Partnerships | Scientific Writing & Publications | NIH Grant Development | Python | R/Bioconductor

## CAREER HIGHLIGHTS

- 18+ years of multidisciplinary biomedical research
- 24 peer-reviewed publications with 700+ citations — Science, Nature Microbiology, The Lancet Infectious Diseases
- Managed ~3,500 clinically annotated patient biospecimens across COVID-19 and sepsis biobanks
- Industry collaborations with Inflammatix, Combinati, and Thermo Fisher Scientific
- Contributed to >$7M in NIH competitive funding
- Mentored 4-5 postdoctoral fellows and physician trainees; participated in postdoctoral recruitment and hiring

## PROFESSIONAL EXPERIENCE

**Basic Life Research Scientist** May 2019 – Present

Stanford University School of Medicine — Department of Emergency Medicine

- Serve as the senior scientific resource supporting multidisciplinary translational research programs, providing leadership in study design, computational biology, quantitative analytics, laboratory operations, regulatory oversight, scientific communications, and external collaborations.
- Design and execute translational research integrating in vitro, ex vivo, and clinically annotated patient biospecimens across infectious disease, immunology, and host-response biology programs.
- Lead and manage COVID-19 and sepsis biobanks comprising ~3,500 clinically annotated biospecimens supporting translational research, biomarker discovery, and industry collaborations.
- Serve as primary scientific liaison for collaborations with Inflammatix and Combinati — coordinating biospecimen utilization, technology evaluation, quantitative analyses, and publications; provided technical feedback to Combinati and Thermo Fisher Scientific on array digital PCR platform performance and research utility.
- Serve as the laboratory's primary scientific reviewer and editor for manuscripts, NIH grant applications, conference abstracts, presentations, and recruitment materials prior to PI review.
- Mentor postdoctoral fellows and physician trainees and participate in interviewing and hiring of postdoctoral researchers.
- Oversee IRB and Administrative Panel on Biosafety (APB) compliance, laboratory operations, procurement, personnel training, and research administration.

- Develop quantitative computational methodologies including machine learning, image analysis, multi-omics integration, and statistical modeling that transform qualitative biological observations into reproducible analytical frameworks.

**Postdoctoral Fellow** Aug 2016 – May 2019

Boston College — Department of Biology

- Investigated bacterial transcriptomics, antimicrobial resistance, and host-pathogen interactions while developing computational approaches and publishing multiple peer-reviewed studies.
- Developed quantitative image analysis and machine learning methods for antimicrobial susceptibility testing; presented findings at ASM, BBM, and CSM national conferences.

**Graduate Research Assistant (Ph.D.)** Aug 2009 – Aug 2016

University of Connecticut — Department of Molecular & Cell Biology

- Conducted doctoral research in microbial genomics, microbial community evolution, and computational biology with international collaborations across multiple institutions.

## SELECTED PUBLICATIONS (24 TOTAL; 700+ CITATIONS)

- Ram-Mohan N., et al. "SARS-CoV-2 RNAemia predicts clinical deterioration and extrapulmonary complications." Clinical Infectious Diseases, 2021.
- Ram-Mohan N., et al. "Using a 29-mRNA host response classifier to detect bacterial coinfections and predict outcomes in COVID-19 patients presenting to the emergency department." Microbiology Spectrum, 2022.
- Haddock NL, Barkal LJ, Ram-Mohan N., et al. "Phage diversity identifies bacterial pathogens in sepsis." Nature Microbiology, 2023.

Full list: Google Scholar | ORCID: 0000-0002-7761-2772

## EDUCATION

**Ph.D., Genetics & Genomics** 2016

University of Connecticut

**B.Tech., Industrial Biotechnology** 2007

Dr. M.G.R. Educational & Research Institute